Prabhavit Gupta



AI Engineer: Agent-Driven Systems, RAG Architectures, Search Pipelines & Production LLM Systems

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Summary

AI Software Engineer with 3 years of hands-on production experience architecting agent-driven systems, enterprise RAG pipelines, and automated retrieval workflows. Proven record designing, shipping, and maintaining production-grade generative AI applications, grounded retrieval engines, and scalable backend infrastructure. Core technical foundation spanning Python (FastAPI, backend services), TypeScript/JavaScript (Node.js, Next.js), PostgreSQL, and cloud deployments, with deep practical specialization in applied generative AI: multi-step agent orchestration, schema-enforced tool calling, vector search and hybrid retrieval, semantic chunking, and automated evaluation loops over live operational logs. Pragmatic builder who operates daily with AI-native coding workflows (Claude Code, Cursor) and prioritizes precision, deterministic verification, and maintainability. Italian resident based in Venice with EU Long-Term Residence Permit (permanent unrestricted work authorization across Italy and the EU without sponsorship); immediately available for Milan hybrid collaboration.

Technical Expertise

- **Agent Architectures & Orchestration:** Multi-step agent workflows, LangGraph, tool and function calling, state management, schema-enforced JSON validation, guardrails, fallback escalation triggers.
- **Search & Retrieval Pipelines (RAG):** Document parsing, semantic chunking, dense vector embeddings, hybrid keyword/semantic search, vector databases, knowledge grounding against dynamic relational datastores.
- **Production LLMOps & Evals:** Automated evaluation pipelines over real customer conversation transcripts, hallucination prevention, latency profiling, token economics, context window optimization, trace observability.
- **Full-Stack & Backend Engineering:** Python (FastAPI, backend services), TypeScript / JavaScript (Node.js, Next.js, React), RESTful APIs, webhook synchronization, automated contract and regression testing (Playwright).
- **Data Systems & Infrastructure:** PostgreSQL, schema design & relational integrity, Supabase, Docker containers, AWS/GCP cloud environments, CI/CD pipelines, Git, workflow orchestration (n8n).
- **Work Authorization & Languages:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted permanent work authorization in Italy, no visa sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer support agent runtime on a live platform serving 4,500+ active users, automating customer inquiries across pricing, venue inventory, and bookings.
- Grounded agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus) using strict JSON tool schemas and fallback escalation logic that routed complex inquiries to staff.
- Built automated evaluation loops analyzing real customer conversation transcripts rather than synthetic benchmarks, systematically detecting failure patterns and eliminating hallucinated answers.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window utilization to reduce inference costs while maintaining rapid response times.
- Deployed automated end-to-end regression test suites using Playwright, verifying API contracts and continuous deployment stability across end-to-end customer booking flows.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting platform growth from 2,000 to 4,500+ users without operational degradation.
- Built automated data ingestion and transformation pipelines parsing and validating unstructured partner listings, expanding platform inventory to over 900 locations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Architected core web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic for client-facing features.
- Implemented automated validation and synchronization pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring relational data integrity.

- Integrated automated CI/CD workflows and regression checks to accelerate feature delivery across distributed development environments.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered bespoke AI automation workflows, web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, fragmented business operations into automated software solutions with structured logging, retry queues, and reliable recovery logic.
- Engineered webhook listeners and real-time synchronization pipelines across CRM platforms, operational tools, and relational datastores.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.